MILITARY SPECIFICATION

EXCAVATORS, DENTAL, CARBON STEEL

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers thin bladed, carbon steel, dental excavators suitable for use in dental operative procedures.

1.2 Classification. Dental excavators covered by this specification shall be of the following type and shape, as specified (see 6.1):

Type I - Black's, Modified, Double Ended

Shape 1 - Hatchet, No. 15, 16
Shape 2 - Discoid, No. 38, 39

Type II - Wedelstaedt's, Modified, Double Ended.

Shape 1 - Hatchet, No. 17, 18

2. APPLICABLE DOCUMENTS

2.1 Specification, standards, and drawing. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

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SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical, General Specifications.
QQ-T-580	Tool Steel, Carbon and Carbon-vanadium.
PPP-B-566	Boxes, Folding Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.

Military

MIL-P-116	Methods of Preservation.
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof Or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center

24288	Excavators, Dental, Carbon Steel.

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 18 Standard Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

ASTM E 92 Standard Method of Test for Vickers Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 General Requirements. The excavators shall be in accordance with the applicable requirements of GG-I-526 and the requirements specified herein.

3.2 Material. The excavators shall be fabricated from water hardening carbon steel drill rod. The steel shall meet the material requirements of QQ-T-580 for Grade A (regular hardening), Class W1-12 1.20 carbon, round drill rod form, spheroidized annealed condition, precision ground and polished finish, for chemical composition, raw material hardness and tensile strength, macrostructure, surface decarburization, carbide form and distribution, hardness penetration and fracture grain size.

3.3 Hardness. Operative ends and shanks of the finished instrument shall have a Rockwell hardness of C56 to C60. Handles shall have a minimum Rockwell hardness of C20.

3.4 Style, design, and dimensions. Excavators shall be one piece, double ended, thin blade design. Type I excavators shall be of Dr. Black's design modified by Dr. Ferrier. Type II excavators shall be Dr. Wedelstaedt's design modified by Dr. Ferrier. The style, design and dimensions of the excavators shall be in accordance with Defense Personnel Support Center (DPSC) Drawing 24288 for the applicable type and shape of excavator. The excavators shall conform to the applicable Black's formula, as specified in the drawing and as specified in Table I herein. Excavators shall be suitable for use in hand excavating and cavity preparations, cleaving or splitting enamel and removal of dentine.

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Table I. Instrument Number and Applicable Black's Formula

Type	Shape	Instrument number	Black's Formula
I	1, Hatchet	15, 16	15-8-14
I	2, Discoid	38, 39	11 1/2-7-14
II	1, Hatchet	17, 18	10-6-14

3.5 Finish.

3.5.1 Operative ends.

3.5.1.1 Type I. The beveled portion shall have a sharp cutting edge. The operative ends (blades) including sharpened edges of the beveled portions shall have a fine honed finish.

3.5.1.2 Type II. The operative ends (blades) shall have a bright finish on the outer face and a fine honed finish on the inner face. The junction of the inner and outer faces (cutting edges) shall form a sharp edge around the periphery of the blade.

3.5.2 Handles and shanks. Excavator handles shall be straight and flat. Handles and shanks of excavators shall be plated as specified in 3.6 and shall have a bright finish. Handles shall have 40 ± 10 fillet or notch-type serrations per inch on seven faces and one smooth face for identification marking. Depth of serrations shall be 0.002 to 0.004 inch.

3.6 Plating. Handles and shanks shall be chromium (nickel over copper undercoat) or nickel over copper plated.

3.7 Corrosion resistance. The plated portions (handles and shanks) of the excavators shall meet the boil test requirements of GG-I-526 when tested as specified in 4.4.2.

3.8 Identification marking. The excavators shall be permanently and legibly marked on the smooth face of the handle. The excavators shall be marked in accordance with GG-I-526 and, in addition, shall be marked with the respective instrument number and Black's formula as specified in Table I. The respective instrument number shall be placed on the handle no less than 5 mm. nor more than 12 mm. from the junction of the handle and shank of the applicable operative end.

3.9 Workmanship. Workmanship shall be first class throughout. The operative end, shank, and handle of the excavators shall be free from defects which may detract from their appearance or impair their serviceability, proper functioning, or intended use.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality for raw material (3.2), supplied by the manufacturer of the steel, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered are classified as specified in Tables II and IV (see 4.2.1 and 4.3) and Table III (see 4.2.2).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in Table II. Unit of product for examination purposes shall be one excavator.

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Table II. Inspection and Acceptable Quality Levels

Type defect	Inspection level	AQL (Percent defective)
Major A	II	1.0
Major B	S-2	1.5
Minor A	II	2.5
Minor B	S-2	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as specified in table III. Lots for hardness testing shall be as specified in 4.2.2.1. The unit of product for inspection purposes shall be one excavator. Acceptance number shall be zero for all sample sizes. Where feasible, the same unit of product shall be used for the determination of two or more test characteristics.

Table III. Sampling Criteria for Tests

Characteristic	Requirement	Test Procedures	Inspection Level
Material	3.2	QQ-T-580	*
Hardness (Finished Instrument)	3.3	4.4.1	S-2
Boil Test	3.7	4.4.2	S-2

* Inspection lots and sampling of raw material shall be in accordance with QQ-T-580.

4.2.2.1 Lots for hardness criteria. For finished instrument hardness testing, a lot or batch of instruments shall consist of instruments produced by one manufacturer at one plant, from the same material (same raw material heat and heat treatment), and from the same heat treatment lot or batch (instruments undergoing heat treatment at the same time, in the same tray, rack or other containing device).

4.3 Examination. The excavators shall be examined to determine compliance with requirements contained in this specification. Examination shall be conducted in accordance with the following classification of defects. Examination shall include, but shall not be limited to, the defects listed in table IV. In addition, examination under magnification shall be as specified in 4.3.1.

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Table IV. Classification of Defects

Categories	Defects
Critical	None defined.
Major A	
101	Style and design for the applicable type, and shape excavator not as specified (see 3.4).
102	Cutting edge of operative end rounded or not sharp.
103	Cutting edge not free from nicks, burrs, feathers, indentations, or other irregularities that affect cutting performance.
104	Operative end finish not as specified.
105	Operative end not free of corrosion.
Major B	
151	Operative end dimensions not as specified.
Minor A	
201	Plating of handle and shank not free from flaking, cracks, or blisters.
202	Plating of handle and shank incomplete or not continuous.
203	Handle and shank finish not as specified.
204	Handle not straight and flat.
205	Identification marking not present, not complete, not properly located, or illegible.
206	Excavators not free from dirt, polishing compound, or foreign matter.
Minor B	
251	Dimensions (other than 151) not as specified on drawing.
252	Handle serrations not uniform or not complete.
253	Handle serration dimensions not as specified.

4.3.1 Examination under magnification. The instruments shall be examined under ten power (10X) magnification for the following number of defects: Categories 102, 103, 104, 105 and 201. Examination under magnification shall include, but shall not be limited to, the defects listed herein.

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4.4 Tests.

4.4.1 Hardness. Finished instrument hardness shall be determined in accordance with ASTM Designation E18 (Rockwell) or E92 (Vickers or Diamond Pyramid).

4.4.2 Boil Test. The excavators shall be subjected to the boil test as specified in GG-I-526.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One excavator of the type specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in the following table shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table V. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	288 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Cleaning and drying. Each unit shall be thoroughly cleaned and dried in accordance with the applicable method as outlined in MIL-P-116.

5.1.3.2 Preservative. Not required.

5.1.3.3 Intimate wrap. Each unit shall be wrapped in neutral tissue or other suitable anticorrosive materials.

5.1.3.4 Unit package. Each unit shall be packaged in accordance with MIL-P-116, method IC-2.

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5.1.3.5 Intermediate package. Intermediate package shall be a box of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except commercial colors will be acceptable, or PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent accidental opening under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4, of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

Note: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636.

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5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages (Military). Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container (Military). Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and shape required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 Limitations. This specification does not cover all types, classes, grades, styles, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover those items which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project Number 6520-1303

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. *Comments and the return of this form* will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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